

Older adults' perspectives on using virtual reality to view nature landscapes in long-term care homes: a qualitative study

Yu-Chen Chi, RN, MSN^{1, }, Hsiao-Ling Hu, RN, MSN^{1, }, Shou-Yu Wang, RN, PhD^{2,3, },
Yeu-Hui Chuang, RN, PhD^{1,4,*, }

¹School of Nursing, College of Nursing, Taipei Medical University, Taipei, Taiwan

²School of Nursing and Midwifery, College of Health, Medicine and Wellbeing, University of Newcastle, Callaghan, New South Wales, Australia

³School of Health, Faculty of Medicine and Health, University of New England, Armidale, New South Wales, Australia

⁴Research Center in Nursing Clinical Practice, Wan Fang Hospital, Taipei Medical University, Taipei, Taiwan

*Address correspondence to: Yeu-Hui Chuang, RN, PhD. E-mail: yeuhui@tmu.edu.tw

Decision Editor: Megumi Inoue, PhD, MSW, RN, FGSA

Abstract

Background and Objectives: Older adults residing in long-term care (LTC) homes (also referred to as LTC facilities) often have limited opportunities to access nature environments, which are known to support psychological well-being. Virtual reality (VR) offers a novel approach to delivering immersive nature experiences for this population. This qualitative study explored the perceptions of using VR to view nature landscapes among older adults in LTC homes.

Research Design and Methods: Using purposive sampling, we recruited 16 older adults (aged 60–84) from 10 LTC facilities in Taiwan. Participants viewed six 360-degree nature videos (each about 2 min long) in a single VR session via a VR headset. Within 1 week of the VR session, participants completed individual, face-to-face, semi-structured interviews. Data were analyzed using content analysis.

Results: Five themes and corresponding categories were generated: (1) as a window to the past, revisiting childhood/youth and reconnecting with places once visited; (2) embracing serenity, feeling relaxed/lightness, savoring the joy, sensing ease/comfort, and feeling inner calm/tranquility; (3) escaping the confines of reality, unlocking a sense of freedom and revisiting natural landscapes; (4) rekindling the healing power of dreams/hope, reigniting hope and motivation and fulfilling unfinished dreams; and (5) unreal and unsettling realities, expressing negative experiences including loneliness, dizziness, and a virtual-reality gap.

Discussion and Implications: Immersive VR nature experiences can evoke nostalgic memories, promote relaxation, provide a temporary sense of freedom, and inspire hope. However, some participants also reported challenges such as dizziness, feelings of artificiality, or loneliness, highlighting the importance of careful design and individualized implementation.

Keywords: Digital nature, Immersive experience, Long-term care, Residents

Innovation and Translational Significance: This study highlights virtual reality (VR) as an innovative and practical approach to address nature deprivation among older adults in long-term care homes. By offering immersive natural landscapes, VR provides a meaningful way for residents to reconnect with the natural world. To facilitate more user-friendly implementation in residential settings, attention should be given to minimizing physical discomfort, enhancing content realism, and mitigating the social isolation that may occur during solitary VR use. These findings can serve as a valuable reference for guiding the future design and implementation of VR interventions in long-term care.

Received: 10 September 2025. **Accepted:** 11 April 2026

© The Author(s) 2026. Published by Oxford University Press on behalf of the Gerontological Society of America.

This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs licence (<https://creativecommons.org/licenses/by-nc-nd/4.0/>), which permits non-commercial reproduction and distribution of the work, in any medium, provided the original work is not altered or transformed in any way, and that the work is properly cited. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com.

Background and objectives

Humans possess an innate inclination to connect with natural environments. [Wilson \(1984\)](#) introduced the biophilia hypothesis, which suggests that this affinity is rooted in our evolutionary development. Building upon this concept, [Kellert and Wilson \(1995\)](#) emphasized that human preference for nature is a fundamental characteristic of our species, influencing our interactions with the environment throughout diverse cultural and historical contexts. According to Attention Restoration Theory ([Kaplan & Kaplan, 1989](#)), exposure to natural environments often contains elements that effortlessly capture attention (soft fascination), allowing the directed attention mechanism to rest and recover from mental fatigue. In addition, viewing unthreatening natural scenes can activate the parasympathetic nervous system, leading to reductions in physiological stress indicators such as heart rate and muscle tension, thereby promoting relaxation ([Ulrich et al., 1991](#)). These perspectives suggest that the human tendency to connect with nature is not only a personal preference but an instinctive drive with significant implications for physical and mental health.

Previous studies have confirmed that exposure to natural environments brings various psychological and emotional benefits for older adults, including enhanced subjective well-being ([Gong et al., 2024](#)), reduced symptoms of depression and anxiety ([Wu et al., 2015](#)), and promotion of positive aging attitudes ([Nikitin et al., 2024](#)). A recent systematic review highlighted that nature-based interventions (NBIs) can further contribute to improving cognitive functioning, increase social interaction, and evoke nostalgia and a sense of life meaning among older individuals ([Tong et al., 2025](#)). Moreover, research has revealed that older adults living in urban areas may experience reduced stress and elevated emotional well-being through access to green and blue spaces, such as parks and waterfronts ([Jarosz, 2023](#)). Prolonged exposure to urban green spaces has also been linked to better physical and mental health ([Lau et al., 2021](#)). Similarly, [Sia et al. \(2020\)](#) reported that 6 months of nature-based activities helped maintain sleep and improved anxiety and cognitive functioning in older adults, highlighting that nature exposure supports both physical and psychological well-being.

When older adults experience aging-related or functional decline that limits independent living, admission to long-term care homes (LTCHs), which offer various levels of care, becomes a common transition. In this study, we adopt the term “homes” rather than “facilities” or “institutions” to reflect a person-centered approach. However, the routines of residential life and residents’ physical limitations often result in significantly fewer opportunities to engage with nature compared to those living in the community. Consequently, NBIs have been gradually introduced into LTCHs in recent years ([Steensma et al., 2025](#)). For example, studies have demonstrated that installing green walls or providing outdoor views within care settings can significantly reduce anxiety and physiological stress indicators, such as residents’ heart rate variability and electrodermal activity ([Xiaoxue & Huang, 2024](#)). [Dahlkvist et al. \(2016\)](#) found that access to therapeutic gardens in LTCHs promotes a sense of relaxation and improves perceived health by fostering “being away” and “fascination,” key elements of nature’s restorative effects. Furthermore, horticultural therapy and outdoor gardens

have been shown to reduce agitation in individuals with dementia, facilitate social interactions, and enhance overall life satisfaction ([Whear et al., 2014](#)). Nature-based approaches have also been linked to improvements in activities of daily living (ADL), well-being, and interpersonal closeness among residents in LTCHs ([Yao & Chen, 2017](#)). A study highlighted the high prevalence of loneliness and depressive symptoms among residents in LTCHs and proposed NBIs as a potential strategy to enhance psychological well-being ([Kolster et al., 2025](#)). Although the growing body of evidence supports NBIs in LTCHs, multiple implementation barriers remain. A recent scoping review identified several obstacles, including residents’ limited mobility, staffing shortages, time constraints, weather conditions, accessibility issues, and infrastructural limitations related to space and safety. These challenges hinder the structural integration of nature elements into care environments and restrict residents’ opportunities for frequent and independent access to outdoor spaces ([Steensma et al., 2025](#)).

In this context, virtual reality (VR) has emerged as an innovative alternative, enabling older adults to experience immersive virtual nature (IVN) even while confined indoors, thus offering comparable psychological benefits ([Kim & Kim, 2025](#)). Although VR-based natural environments provide opportunities for residents to virtually reconnect with nature, limited research has explored how older adults in LTCHs perceive and experience these technologies.

[Muslu et al. \(2025\)](#) conducted interviews with community-dwelling older adults regarding their experiences with VR “time travel” scenarios, which focused on reminiscence-based immersive environments rather than nature-specific content, and found that participants felt as though they could “go back in time,” reliving meaningful life events that evoked the feelings of happiness, joy, and nostalgia. Similarly, [Kim and Kim \(2025\)](#) investigated the psychological effects of IVN among older adults living in longevity villages. Their mixed-method study concluded that IVN fostered positive emotions, such as gladness and happiness, reduced stress, and alleviated feelings of depression. In addition, [Appel et al. \(2020\)](#) conducted a feasibility study across multiple care settings, including a Canadian day treatment center, a rehabilitation hospital, and a long-term care facility, involving older adults with varying levels of cognitive impairment, including dementia, as well as physical disabilities. The study found that 76% of older adults expressed willingness to use VR again after viewing 360-degree nature scenes, and no negative side effects were reported. Importantly, the feasibility and acceptability of the VR intervention were similar across all three settings. Similarly, [Hosseini et al. \(2024\)](#) investigated the acceptability of VR-based games among older adults in LTCHs, revealing that participants found the technology engaging and motivating, with potential benefits for promoting physical activity and social connection; however, the intervention did not involve virtual natural environments.

Overall, previous studies suggest that VR is an applicable tool for use among older adults. However, existing VR studies have largely been conducted in community-based settings, while research within LTCHs has primarily focused on technical feasibility or non-nature-based applications. This has left a gap in our understanding of how residents in LTCHs perceive the use of VR to visit natural environments. Although physical discomforts

such as dizziness, nausea, and eyestrain have been reported during immersive VR sessions (Healy et al., 2022; Schaumburg et al., 2025), a systematic review by Drazich et al. (2023) concluded that the overall benefits of VR interventions outweigh the risks associated with mild cybersickness. These findings suggest that research could move toward exploring the subjective perspectives of residents regarding virtual nature. Therefore, this study aimed to explore the perceptions of the use of VR to view natural landscapes among older adults in LTCHs. The findings are expected to inform the future application and design of VR-based natural interventions.

Research design and methods

Research design

This study used a qualitative research design and was reported in accordance with the Standards for Reporting Qualitative Research (SRQR) guidelines (O'Brien et al., 2014). This study was part of a larger project on virtual nature in LTCHs.

Research setting and recruitment

Participants were recruited via purposive sampling from 10 LTCHs, encompassing a range of care levels, including one nursing home, seven long-term care institutions, and two assisted living facilities. As mentioned previously, these settings are referred to as “homes” to reflect a person-centered perspective. Recruitment continued until data saturation, which indicates that no new codes or themes emerged (Hennink et al., 2017), resulting in a total of 16 participants in the study. There were no refusals or withdrawals among the participants throughout the study.

Inclusion and exclusion criteria

The inclusion criteria were the older adults who (1) were aged 60 years or above, (2) had resided in an LTCH for at least 1 month, (3) were cognitively intact and able to articulate personal experiences, and (4) were willing to share their experiences. The exclusion criteria included the older adults who (1) were blind, (2) had infectious diseases, (3) had conditions that substantially impaired visual function, such as advanced stages of cataracts, glaucoma, or macular degeneration, and (4) reported susceptibility to motion sickness, light sensitivity, or claustrophobia.

Data collection

After receiving approval from the Institutional Review Board (IRB), one of the researchers contacted LTCHs by phone and explained the study's purpose and procedures to the head nurses or facility managers. To ensure participants were cognitively intact, a two-stage screening process was employed. First, staff in LTCHs pre-screened residents based on available medical records to identify those without a documented diagnosis of dementia. Subsequently, the researcher confirmed cognitive status using the Short Portable Mental Status Questionnaire (Pfeiffer, 1975). Only those with an error score of ≤ 2 (indicating

intact cognitive function) were included in the study. Written informed consent was obtained from each participant who agreed to participate in the study before data collection.

After obtaining informed consent, participants used the HTC VIVE Flow immersive VR headset. Its lightweight design (189 g) was specifically chosen to ensure comfort for older adults in LTCH settings. Participants viewed six 360-degree VR nature videos (forest, grassland, mountain, lake, river, and ocean) once each during a single VR session. All videos were recorded in 4K resolution with accompanying natural ambient sounds (e.g., birdsong, flowing water, wind) and did not include narration. The videos were professionally recorded by licensed videographers and obtained with authorization for research use. To minimize the risk of sensory conflict or dizziness, all stimuli featured in-scene movement with a static camera, filmed from a fixed-point perspective (tripod-mounted). There was no forward motion, panning, or simulated locomotion, thereby ensuring a stable visual frame while maintaining dynamic environmental content, such as breaking waves or leaves moving in the wind. Each video lasted approximately 2 min, and the six videos were presented in a randomized order. Moreover, the researcher remained present during each VR session to assist with headset placement and monitor for any signs of discomfort. To maintain procedural consistency and minimize researcher-induced bias, a non-intrusive approach was adopted; the researcher interacted with participants only when immediate physical assistance or technical support was required. This protocol was designed to ensure that the reported perceptions regarding immersion were reflective of the VR content, rather than social interaction between the participant and the researcher.

To ensure the reliability of participant reflections, individual face-to-face, in-depth interviews were conducted within 1 week of the VR session using a semi-structured interview guide by the same female researcher who accompanied the participant. This researcher possesses 1 year of qualitative research experience and 6 years of clinical experience in geriatric and long-term care. Throughout the study, the research team maintained a reflexive stance. They explicitly acknowledged a pre-existing assumption that immersive VR would serve as a primarily restorative escape. To mitigate this bias, the team utilized a semi-structured interview guide with open-ended questions, reflexive journaling, and peer debriefing to ensure the findings remained grounded in the residents' lived experiences.

The semi-structured interview guide included the following questions: (1) Can you describe your experience of viewing natural landscapes through VR? (2) Which types of virtual natural landscapes do you prefer, and why? (3) What do you see as the advantages and disadvantages of using a VR headset to view natural landscapes? (4) Do you have suggestions for improving the VR experience? (5) How did you feel during the experience of viewing natural landscapes using VR? (6) Is there anything else you would like to share? During the interviews, clarification probes such as “Can you tell me more?”, “Do you mean...?”, “What were you thinking at that moment?”, and “Can you give an example?” were used to encourage participants to think deeply. Interviews were conducted in private and quiet settings, either in a discussion room or the participant's room, based on their preferences, with only the participant and the researcher present. All interviews were audio-recorded using a digital

recorder and transcribed verbatim within 72 h. In addition, the researcher took field notes documenting participants' facial expressions, body language, and the researcher's reflections. Each participant was interviewed once, and each interview lasted an average of 51.25 min ($SD=9.27$). Demographic and background information, including age, gender, education, marital status, religion, ADL, and length of stay in the LTCH, was collected after the interview. Data collection took place from October 2023 to June 2024.

Data analysis

This study employed content analysis (Graneheim et al., 2017) to analyze the qualitative data. The analysis process involved several steps: first, the interview transcripts were read multiple times to ensure deep familiarity. Next, meaning units were identified and condensed into shorter meaning units. These were then assigned codes, which were grouped into categories based on shared characteristics. Finally, these categories, which had similar meanings, were further abstracted into themes. Field notes documenting participants' facial expressions, body language, and researcher reflections were integrated with the transcripts during coding to enrich contextual interpretation. Two researchers independently conducted the data analysis, with discrepancies resolved through consensus. An example of how initial codes were developed and grouped into themes is presented in Figure 1.

Rigor of the study

To ensure the rigor of this study, four criteria proposed by Lincoln and Guba (1985) were applied: (1) Credibility: All

researchers had formal training in qualitative research. Member checking was conducted with two participants to verify the accuracy of the findings, and peer debriefing with experienced qualitative researchers was performed to enhance the study's credibility. (2) Dependability: To ensure consistency, all interviews were conducted by the same researcher, transcribed verbatim within 72 h, and cross-checked for accuracy. All original data were securely stored, and the entire research process was thoroughly documented. (3) Transferability: During the interviews, participants were encouraged to share detailed descriptions of their experiences. In addition, participants were recruited from various LTCHs, and contextual information about the settings and participants was thoroughly documented to enhance the transferability of the findings. (4) Confirmability: During interviews, the researcher actively verified participants' responses. In addition, two researchers independently analyzed the transcripts and jointly discussed and reviewed the identified themes and categories to ensure objectivity. All raw data were retained in full for potential future review and validation.

Ethical considerations

This study was approved by the Taipei Medical University–Joint Institutional Review Board (TMU-JIRB) (Approval No. N202304015; Approval date: April 20, 2023) and was conducted with the permission of participating LTCHs. Written informed consent was obtained from all participants, whose rights and well-being were fully protected. All quotes were labeled with generic identifiers, such as “Participant A,” to ensure anonymity and reduce the risk of indirect identification.

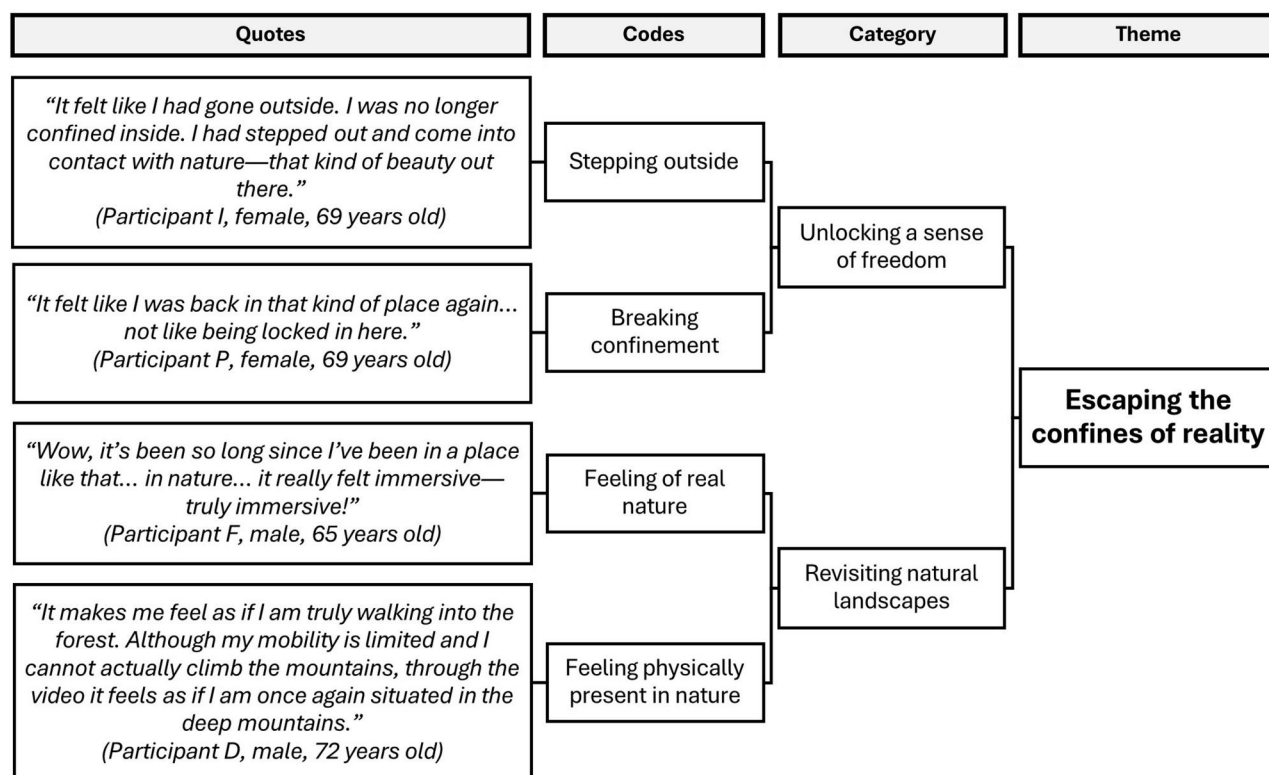


Figure 1 Example of the data analysis process.

Results

Demographics and characteristics of participants

The 16 participants had a mean age of 73.3 years ($SD=8.4$; range: 60 to 84), and the majority were female (62.5%). Most participants (81.3%) completed senior high school or above. Physical functional status was assessed using the Barthel Index, with an average score of 76.9 ($SD=35.1$), where higher scores indicate greater independence in ADL (range: 0–100) (Table 1).

Perspectives on using VR to visit nature landscapes among the older residents in LTCHs

Five themes were generated: (1) As a window to the past, (2) Embracing serenity, (3) Escaping the confines of reality, (4) Rekindling the healing power of dreams and hope, and (5) Unreal and unsettling realities (Table 2).

As a window to the past

Nine of the 16 participants described this experience. The virtual nature landscapes evoked personal memories and emotional resonance among older residents in LTCHs, reinforcing connections to their life histories. These immersive scenes not only offered a sense of emotional warmth but also brought back vivid recollections of earlier life stages. Two categories were identified: revisiting childhood and youth, and reconnecting with places once visited.

Revisiting childhood and youth

Certain virtual environments triggered memories from participants' childhood or adolescence, enabling them to relive past

experiences and emotional states. These nostalgic connections enhanced their sense of personal continuity.

I felt very touched... deeply touched... These seascapes reminded me of my childhood home in Penghu. I'm from Penghu. It felt so familiar—like my hometown. (Participant C, female, 65 years old)

I like the ocean. I used to go swimming a lot when I was young... It reminded me of my school days—not really childhood, more like high school. Those were happy times ... Both, actually! Seeing the vast ocean made me happy, and the scene reminded me of my youthful high school years—it was joyful. (Participant N, female, 81 years old)

Reconnecting with places once visited

Some virtual scenes resembled places the participants had visited in the past, either during travels or as part of their daily routines. These familiar elements evoked emotional connections tied to specific locations and life experiences.

I used to live in Taipei City, near Rongxing Garden. I went to the park every morning and walked there every afternoon. I did that for over 20 years, so I really liked the plain. (Participant E, male, 84 years old)

Sometimes it's like reliving old memories... revisiting old dreams. Ah, yes, Canada has that clock that goes ding-dong-ding-dong... It brings back memories of visiting that place. (Participant K, female, 82 years old)

Embracing serenity

Most participants (13 of the 16) described this experience. The IVN experience provided both psychological and physiological relaxation for older residents, serving as a meaningful medium to reduce stress and soothe emotions. During the VR sessions,

Table 1 Demographics and characteristics of participants ($N=16$).

Participant	Gender	Age (years)	Education level	Barthel index
A	Female	83	College	100
B	Female	70	Senior high school	100
C	Female	65	Senior high school	25
D	Male	72	Primary school	85
E	Male	84	Senior high school	95
F	Male	65	Junior high school	15
G	Male	62	College	85
H	Female	60	College	95
I	Female	69	Senior high school	90
J	Male	67	Senior high school	40
K	Female	82	Senior high school	100
L	Female	78	College	100
M	Male	83	College	100
N	Female	81	College	100
O	Female	82	Senior high school	100
P	Female	69	Junior high school	0

Note. Barthel Index (range: 0–100); higher scores indicate greater independence in activities of daily living.

Table 2 Themes and categories reflecting older residents' perspectives on virtual nature landscapes.

Theme	Category
1. As a window to the past	(1) Revisiting childhood and youth (2) Reconnecting with places once visited
2. Embracing serenity	(1) Feeling relaxed and lightness (2) Savoring the joy (3) Sensing ease and comfort (4) Feeling inner calm and tranquility
3. Escaping the confines of reality	(1) Unlocking a sense of freedom (2) Revisiting natural landscapes
4. Rekindling the healing power of dreams and hope	(1) Reigniting hope and motivation (2) Fulfilling unfinished dreams
5. Unreal and unsettling realities	(1) Feeling loneliness and isolation (2) Feeling dizziness (3) Virtuality–reality gap

participants gradually entered a state of inner balance, tranquility, and contentment. Four categories were identified: feeling relaxed and lightness, savoring the joy, sensing ease and comfort, and feeling inner calm and tranquility.

Feeling relaxed and lightness

Many participants reported a sense of release and relief in both body and mind, resulting in a feeling of lightness and relaxation during the virtual nature experience.

Viewing nature is the best for us. It gives you a kind of cleansing for both body and mind. It makes you feel free from all the worldly troubles—you can just relax. (Participant A, female, 83 years old)

In terms of how it feels—psychologically. If someone tends to feel a bit depressed, I think watching this would help them feel more relaxed. (Participant L, female, 78 years old)

Savoring the joy

The virtual landscapes elicited positive emotions such as joy, delight, and satisfaction. These emotionally uplifting moments provided participants with a sense of renewal and happiness.

Just seeing such beautiful scenery is enough. Simply appreciating this beautiful landscape—nature—makes me feel happy. (Participant J, male, 67 years old)

It felt like a miracle of life. As if I had gone on a scenic tour ... I felt so refreshed, so uplifted. When I saw the green trees, I felt truly happy. (Participant K, female, 82 years old)

Sensing ease and comfort

Specific elements within the nature landscapes, such as greenery, flowing water, and expansive views, contributed to an overall sense of comfort and evoked pleasant physical and emotional sensations.

I really enjoy watching those wide, boundless views. They make me feel so comfortable, like my whole body feels calm and refreshed. (Participant B, female, 70 years old)

The sound of streams and the sound in forests are different, but both are refreshing. With the river, trees, and birdsong, it felt like listening to a symphony—it was very soothing. (Participant E, male, 84 years old)

Feeling inner calm and tranquility

Participants described a profound sense of emotional stabilization and inner peace brought about by the immersive nature experience. The tranquil atmosphere contributed to emotional regulation and inner stillness.

When I see the ocean, it calms my mind. The colors of the sea make me feel comfortable, peaceful, and happy. It really helps me stop thinking about anything negative. (Participant H, female, 60 years old)

We need to open our hearts. That's the main point—this video ... it really helped to calm and stabilize my heart, especially for someone who's ill. It helps with the illness. The key is to find calmness in the heart—then the body will improve. (Participant J, male, 67 years old)

Escaping the confines of reality

For older adults residing in LTCHs, IVN experiences offered a form of psychological “escape” and a renewed sense of freedom. Eight of the 16 participants described this experience. While viewing the VR landscapes, participants experienced a temporary detachment from illness, functional impairments, and the monotony of residential routines, leading to a subjective feeling of liberation and autonomy. Two categories emerged: unlocking a sense of freedom and revisiting natural landscapes.

Unlocking a sense of freedom

The virtual nature experiences gave participants the impression of “leaving the facility” and freely accessing outdoor environments. This fostered a psychological sense of mobility and autonomy despite physical limitations.

It felt like I had gone outside. I was no longer confined inside. I had stepped out and come into contact with nature—that kind of beauty out there. (Participant I, female, 69 years old)

It felt like I was back in that kind of place again ... not like being locked in here. (Participant P, female, 69 years old)

Revisiting natural landscapes

Some participants expressed that the virtual experience felt like a return to nature. The immersive quality of the VR environment created a strong sense of presence, as if they were physically back in those natural spaces.

Wow, it's been so long since I've been in a place like that ... in nature ... it really felt immersive—truly immersive! (Participant F, male, 65 years old)

It makes me feel as if I am truly walking into the forest. Although my mobility is limited and I cannot actually climb the mountains, through the video, it feels as if I am once again situated in the deep mountains. (Participant D, male, 72 years old)

Rekindling the healing power of dreams and hope

In addition to evoking memories of the past, the virtual nature experiences also sparked imagination and inspired hopes for the future. The visual and emotional stimulation reignited participants' inner desires and intentions for action, highlighting the potential of virtual environments to enhance psychological resilience and renew life's meaning. Eight of the 16 participants described this experience. Two categories were identified: reigniting hope and motivation and fulfilling unfinished dreams.

Reigniting hope and motivation

Some participants reported that viewing virtual nature scenes uplifted their mood and even inspired thoughts of going outdoors, participating in rehabilitation, or traveling again. The VR experience served as a catalyst, shifting them from passive observation to active reflection of future possibilities.

We long for it. It makes you hope that one day you can actually go there. We don't want to lie in bed all day. We don't want just virtual experiences—we want to go out for real. (Participant K, female, 82 years old)

Watching this made me feel comfortable. It gave me something to look forward to. Although you can't really take us there, this kind of experience makes me think that maybe one day I will go (in person). (Participant A, female, 83 years old)

Fulfilling unfinished dreams

For some residents, the virtual experience offered a way to emotionally connect with places they had never visited. These scenes served as symbolic representations of lifelong aspirations, allowing participants to feel as though they had “been there,” and providing a sense of emotional closure or fulfillment.

It's been so long since I've seen places like that. Watching them—even as pictures or videos—still satisfies some of my desires. (Participant O, female, 82 years old)

I've talked about it many times, but I've never done it. Through this opportunity, I felt like I realized it for a moment—at least psychologically. If it weren't for you, I might never have the chance to go there in this lifetime... It fulfilled my dream. It felt like I was really there again. (Participant C, female, 65 years old)

Unreal and unsettling realities

While many participants responded positively to the IVN experience, some expressed emotional or physical discomfort, such as feelings of loneliness, dizziness, or a sense of artificiality. These reactions highlight individual differences in adaptability and the limitations of VR technology for older adults. Eight of the 16 participants described this experience. Three categories were identified: feeling loneliness and isolation, feeling dizziness, and the virtuality–reality gap.

Feeling loneliness and isolation

Some participants reported heightened feelings of loneliness during the virtual experience, particularly due to the lack of social interaction. Watching nature scenes alone reminded them of their real-life isolation, as the virtual setting did not simulate the presence of companions.

It felt just like when I look outside every day—I feel lonely. Watching this, I wondered why I was the only one viewing

it. I couldn't find anyone to interact with (when visiting the scenes). (Participant G, male, 62 years old)

... Watching it alone felt lonely... Having someone there is more important. (Participant D, male, 72 years old)

Feeling dizziness

A few participants experienced dizziness or visual discomfort while using the VR headset, which interrupted their sense of immersion and made it difficult to fully engage with the experience.

The disadvantage was the shaking. For me, it was shaky, and I felt really dizzy. You're not just looking straight ahead, right? When it moved (the ocean waves), my head moved with it—it made me dizzy. (Participant A, female, 83 years old)

I felt like there was something wrong with my eyes. When the images were too close (to my eyes), I got a bit dizzy. Only at close distance though. Then I felt a little dizzy. (Participant D, male, 72 years old)

Virtuality–reality gap

Although the VR experience provided visual stimulation, some participants noted the absence of multisensory cues, such as wind, fresh air, and natural scents, which diminished the sense of authenticity. In addition, wearing the headset also contributed to a feeling of detachment.

There's no ocean breeze when you're at the beach, no mountain wind or fresh air when you're in the mountains. You're just indoors looking at it—appreciating it—but it's not like actually being at the seaside or on the grassland. That feeling... that experience isn't the same. (Participant M, male, 83 years old)

Nature... it's just an image they give you. It's completely different from actually being there. This is just a headset showing you images. If you were truly there, it would feel entirely different. (Participant L, female, 78 years old)

Discussion and implications

This study explored the subjective experiences of older residents in LTCHs who engaged with immersive VR to view natural landscapes. The findings indicate that, despite physical limitations and residential confinement, VR can help older adults overcome both tangible and intangible barriers to nature. By enabling residents to reconnect with natural environments, our results highlight the potential value of implementing virtual nature interventions within the LTCH context. However, the experience was also characterized by a loneliness paradox; while the technology offered a digital escape from residential confinement, it simultaneously highlighted a sense of social isolation during solitary immersion. This suggests that the psychological impact of

VR is not purely restorative but is intricately shaped by the residential and social context of long-term care. Furthermore, some participants noted physical challenges, such as dizziness or a perceived lack of realism in the virtual imagery, which must be addressed to optimize future implementation.

The first theme, “as a window to the past,” revealed that participants spontaneously recalled personal memories, such as “places I used to visit as a child,” while viewing virtual nature scenes. These memories were often emotionally rich and tied to earlier life stages. This aligns with findings by [Yan et al. \(2023\)](#), who identified nostalgia as a mediator in the emotionally restorative effects of natural environments. In other words, nature scenes may promote emotional well-being by stimulating nostalgic recollections, which in turn foster relaxation, uplift mood, and provide psychological comfort.

Interestingly, previous VR-based reminiscence interventions have often relied on personalized materials, such as family photos, images of former homes, or personal life events to trigger nostalgic responses. A systematic review by [Mao et al. \(2024\)](#), which included 22 studies focusing on older adults with cognitive impairments, found that most interventions used individualized stimuli as the basis for evoking reminiscence. The review highlighted that personalized content is one of the key factors influencing the effectiveness of VR-based reminiscence interventions. However, the natural scenes used in both [Kim and Kim \(2025\)](#) and our study were not personalized, yet participants still associated them with meaningful life events, hometowns, and childhood experiences. This indicates that even non-personalized nature scenes can evoke autobiographical memories and nostalgic emotions among older adults. The inherent features of natural landscapes may be sufficient to elicit powerful emotional responses. Moreover, this phenomenon can be theoretically situated within Kaplan’s Attention Restoration Theory (ART) ([Kaplan & Kaplan, 1989](#)). The “soft fascination” and sense of “being away” provided by nature scenes allow the mind to disengage from the “directed attention” demands of daily institutional life. Under such low-demand attentional states, [Smalley et al. \(2023\)](#) suggest that virtual nature may function as a cue for involuntary autobiographical memory activation, in which personal memories spontaneously surface without conscious effort. These findings support the potential of nature-based VR interventions as a practical and scalable option in LTCHs, particularly where individualized content or direct access to real nature is limited.

Second, the theme “embracing serenity” reflects the sense of relaxation, happiness, ease, and inner calm experienced by residents during the virtual nature sessions. Many participants described their experiences using phrases such as “relaxed in both body and mind,” “calm and stable,” and “refreshed and uplifted,” suggesting that IVN can effectively promote emotional relief and psychological comfort. This finding is consistent with [Appel et al. \(2020\)](#), who conducted a VR nature experience study among older adults. Although primarily quantitative, their study also included qualitative feedback, for example, “*I picture myself being there to relax*” ([Appel et al., 2020](#), p. 8), indicating that virtual nature can elicit a clear sense of relaxation. In addition, the study by [Kim and Kim \(2025\)](#) found that VR-based natural environments can alleviate stress and feelings of depression and can enhance gladness, happiness, and mental relaxation

among older adults. These observations can be further understood through [Ulrich’s \(1984\)](#) Stress Reduction Theory (SRT), which posits that exposure to natural environments can reduce physiological and emotional arousal by regulating sympathetic nervous system activity, thus promoting psychological recovery and relaxation. While Ulrich’s original theory was based on real-world exposure to physical nature, the current findings suggest that even non-physical, indoor-based virtual nature formats may offer similar emotion-regulating benefits. Future studies are encouraged to evaluate this effect further using large-scale quantitative approaches.

Although the virtual nature videos used in this study lasted only 2 min, most participants still reported feeling relaxed and psychologically soothed. This finding is consistent with previous research by [Yuan et al. \(2022\)](#), which demonstrated that even a brief exposure (just 3–5 min) to virtual forest environments can lead to noticeable psychological relaxation and emotional improvement. In summary, our findings not only support Ulrich’s theoretical perspective on stress recovery but also resonate with the Biophilia Hypothesis, which suggests that humans possess an innate emotional connection to nature. Even when nature is experienced through virtual representations, it may foster feelings of familiarity and connection that contribute to therapeutic benefits.

Furthermore, this study found that older adults perceived IVN as a way of “escaping the confines of reality.” Residents in LTCHs often face spatial constraints and limited physical mobility, resulting in restricted daily environments and dependence on caregivers for outdoor access. The findings suggest that virtual nature experiences not only simulate natural environments through visual stimuli but also offer a form of psychological displacement and symbolic detachment from daily residential life. Several participants described the experience as feeling like they had “left the long-term care facility,” were “not locked in here again,” or had “mentally stepped away from this place.” These expressions reflect the immersive feature of VR, which, as noted by [Bauer and Andringa \(2020\)](#), can induce a strong sense of presence that allows users to temporarily dissociate from their physical surroundings and feel as though they are situated within the virtual environment.

This phenomenon aligns with the concept of “being away” in Kaplan’s ART, which refers to a psychological sense of distance from routine tasks and stressors that facilitates cognitive restoration. However, in the context of virtual nature, “being away” is inherently symbolic and psychological. It does not require physical relocation from the LTCH but instead enables a temporary mental escape, evoking a fleeting sense of freedom and imagined mobility. Some participants commented, “It’s been a long time since I’ve seen such natural scenery... it felt like being there in person,” or “I felt like I really went to that place.” These immersive experiences appeared to allow residents to temporarily transcend the constraints of residential living, fostering a psychological sense of liberation and symbolic departure.

Some older adults described the virtual natural landscapes as “rekindling the healing power of dreams and hope,” highlighting their potential to serve as a motivational force that fosters a positive and future-oriented mindset. Several participants made remarks such as “I want to really go see it,” “I want to travel again,” or “this gives me something to look forward to.” Field

notes corroborated that these statements were accompanied by smiles, brightened eyes, and forward-leaning postures, suggesting eagerness and anticipation. These expressions indicate that virtual nature not only alleviates emotional distress but also acts as a catalyst for reawakening the desire for active engagement with the world. Such experiences may inspire new goals or life directions that hold particular meaning for residents who have long been confined to more passive and dependent roles in residential care settings. Virtual nature may thus serve as a transformative medium for reigniting unfulfilled dreams and renewing a sense of hope. Similarly, Muslu et al. (2025) also reported that virtual nature experiences enabled some older adults not only to recall cherished memories but also to express a renewed desire for future travel. Kim and Kim (2025) found that some older adults, after engaging with virtual nature, reported “feeling uplifted,” with some expressing intentions to travel again. These findings suggest that virtual nature landscapes can reignite a sense of vitality and motivation, encouraging older adults to pursue healthier, more active living in the hope of one day revisiting real nature. Overall, these findings point to a coherent psychological process through which immersive VR nature supports emotional restoration, autobiographical meaning-making, and future-oriented motivation. Drawing on principles from ART and SRT, IVN may reduce attentional demands and promote emotional calming, creating conditions in which autobiographical memories and nostalgic meanings emerge and may extend toward renewed hope and motivation.

Although the current study found that virtual nature experiences generally elicited positive responses, some older adults expressed feelings of “unreal and unsettling realities,” highlighting the limitations and challenges in such interventions. Some participants remarked that VR still felt artificial and could not fully replicate the authenticity of real nature, leading to comments such as “it doesn’t feel real” or “I still feel like I’m indoors.” Compared to real natural environments, VR remains limited in sensory fidelity, particularly due to the absence of tactile sensations, scents, and other multisensory elements that characterize real-world nature. Future technological enhancements, especially the integration of touch, smell, and even taste alongside visual and auditory stimuli, are recommended to create more immersive and hyper-realistic experiences.

Another reported issue was dizziness. While not experienced by all participants, some described discomfort during VR use. Although these symptoms typically subsided after a brief rest, they remain a concern. Leung et al. (2022), in a systematic review, addressed that older adults may experience cybersickness during immersive VR, including symptoms such as dizziness and nausea. Similarly, Healy et al. (2022) found that rapid visual transitions or dynamic scenes can intensify VR-induced motion sickness in older populations. These findings suggest that minimizing motion-related discomfort should be a main design consideration for future VR applications targeting older adults.

Most importantly, an additional concern raised by some participants was a sense of isolation. Comments such as “watching VR alone feels lonely” suggest that the absence of social interaction in VR nature experiences may diminish emotional engagement for some older adults, and could even intensify feelings of loneliness. Beyond a simple lack of company, the virtual world reminds residents of the social and physical boundaries they

face in their daily lives at LTCHs, making those constraints feel more intense. This phenomenon aligns with the Technology and Consumer Well-being Paradox Model (McLean et al., 2025), which identifies technology as a “double-edged sword.” While VR is intended to enhance well-being by providing a sensory escape to nature (a positive benefit), it paradoxically triggers a negative well-being state by exposing the gap between the perceived richness of the virtual environment and the participant’s actual social isolation within the residential setting.

As discussed earlier, the experience of “being away” involves a sense of psychological distance that allows individuals to disengage from routine demands. However, this psychological distancing was not universally achieved. In highly structured long-term care environments, where daily life is shaped by residential routines, prolonged waiting, and limited opportunities for social interaction, IVN may fail to create sufficient separation from residents’ ongoing lived realities. Under these conditions, the solitary consumption of nature, which is often a socially shared experience, creates a form of “emotional dissonance.” For some older adults, immersive exposure may resonate with existing experiences of social confinement rather than offering psychological relief. Consistent with this interpretation, Finnegan and Campbell (2023) noted that older adults who experience temporary social satisfaction in VR may feel a greater sense of emotional emptiness upon returning to a reality devoid of such interactions, thus exacerbating feelings of isolation. Importantly, this paradoxical response is not unique to immersive VR but reflects a broader challenge observed across technology-based interventions targeting loneliness in later life. A meta-analysis by Shah et al. (2021) found no consistent evidence that digital technology interventions effectively reduce loneliness among older adults, underscoring that digital connection does not necessarily translate into meaningful social connection. Similarly, Balki et al. (2022) reported that technology-based interventions are more likely to be beneficial when they actively facilitate engagement with existing social networks rather than function as passive or substitutive experiences. These findings suggest that immersive virtual experiences, when implemented without opportunities for interpersonal engagement, may inadvertently intensify feelings of isolation, particularly in residential settings where social deprivation already exists. The present findings extend this literature by illustrating how such paradoxical emotional responses are experienced and interpreted by older adults living in LTCHs when engaging with IVN. Consequently, IVN is better understood not as an inherently restorative intervention but as one whose impact depends on whether environmental and social conditions support the activation of psychological “being away.” Crucially, this finding suggests that ART, while effective in explaining individual restorative processes, does not fully account for the social-emotional dynamics that emerge when immersive technologies mediate nature experiences in socially constrained residential settings.

Accordingly, when IVN alone is insufficient to support the experience of “being away,” interventions should move beyond treating VR as a passive substitute for social interaction. Prior research on social VR indicates that meaningful social engagement in immersive environments does not emerge spontaneously but depends on structured interaction design and active facilitation (Kalantari et al., 2023). Therefore, we recommend reframing

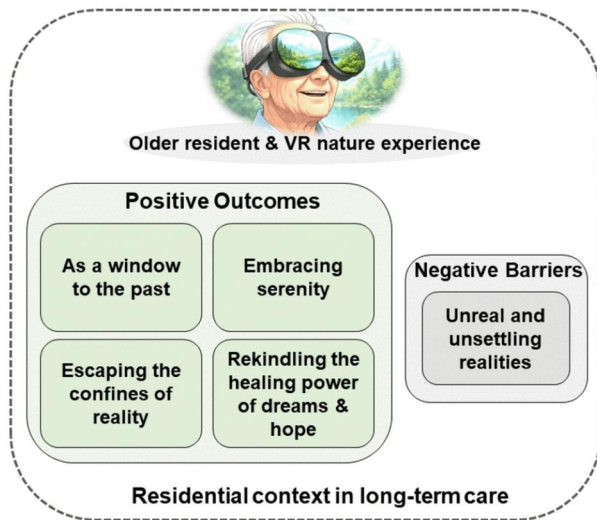


Figure 2 A conceptual diagram representing the five themes generated from interviews with older residents. Note. VR = virtual reality. Themes are situated within the residential long-term care context and are grouped into positive outcomes and negative barriers associated with using VR to visit natural environments.

immersion as a socially situated experience. This involves offering VR in paired or small-group settings, supplemented by “social anchoring” (such as staff accompaniment) and followed by guided reflection to facilitate shared processing and group reminiscence (Stefan et al., 2025). Rather than a self-contained activity, such structuring maintains relational continuity while supporting psychological disengagement. This integrated approach aims to mitigate the “emotional dissonance” caused by the contrast between virtual beauty and physical isolation, ensuring that the healing power of nature effectively aligns with the complex social needs of LTCH residents.

Figure 2 provides a conceptual synthesis of these findings, illustrating the perceptions of older residents regarding the use of VR to visit natural environments.

Strengths, limitations, and future directions

This study provides rich perspectives from older residents across 10 LTCHs, enhancing the diversity and transferability of the findings. However, several limitations must be acknowledged. First, this study used purposive sampling to recruit the residents who were willing to share their experiences; consequently, the findings may not fully reflect the perspectives of those who were less inclined to participate in research or less open to technology. Second, we did not investigate participants’ nature affinity or personal history with natural environments in this study. Future research should consider including this background information to better understand the role of nature-relatedness in VR experiences. Third, while 360-degree videos provide immersion, the limited realism and lack of interactivity in static video content should be noted. To strengthen immersion in future studies, researchers may incorporate multisensory modalities such as tactile feedback, olfactory cues, and wind

simulation. Future research is also needed to explore strategies for minimizing dizziness and cybersickness, including the optimization of visual transitions and scene pacing. Moreover, exploring shared VR experiences may alleviate loneliness by offering residents a “socially structured escape,” allowing them to move beyond their daily routines together through collective immersion. From a practical perspective, VR nature interventions in long-term care settings could be made feasible by embedding them within existing recreational routines. In addition, rather than relying solely on busy nursing staff, these interventions could be supported by activity coordinators, trained volunteers, or family members. Finally, future studies should build on the current findings by adopting longitudinal or mixed-method designs. Longitudinal approaches could examine the durability of VR-related psychological benefits over time, while mixed-method approaches combining qualitative perspectives with physiological measures (e.g., heart rate variability or cortisol levels) would allow for a more comprehensive assessment of the sustained psychological impact of VR nature interventions.

Conclusion

This qualitative study explored the perceptions of older residents in LTCHs who used VR to view natural landscapes. Five themes emerged from the data: as a window to the past, embracing serenity, escaping the confines of reality, rekindling the healing power of dreams and hope, and unreal and unsettling realities. The findings suggest that IVN experiences may help older adults evoke meaningful personal memories; experience relaxation, happiness, comfort, and tranquility; gain a sense of psychological freedom; and inspire renewed motivation and hope for the future. While participants generally responded positively, some also expressed concerns, such as the artificial nature of the experience, dizziness, and feelings of loneliness. These challenges suggest important considerations for future VR design and content development. Overall, this study highlights the potential of virtual nature as a feasible and supportive intervention in LTCHs, especially for residents with limited access to real natural environments. However, careful attention to individual responses, preferences, and safety is essential to ensure that these experiences are both effective and ethically implemented in residential care settings.

Funding

This research was supported by a grant from the National Science and Technology Council (NSTC), Taiwan (grant no.: 112-2314-B-038-026-MY3).

Conflicts of interest

None declared.

Data availability

Data available on request from the authors. The study was not preregistered.

Author contributions

Yu-Chen Chi: Conceptualization, Methodology, Formal analysis, Investigation, Writing—Original draft. Hsiao-Ling Hu: Conceptualization, Methodology, Formal analysis. Shou-Yu Wang: Conceptualization, Methodology. Yeu-Hui Chuang: Conceptualization, Methodology, Validation, Resources, Data curation, Writing—Original draft, Supervision, Funding acquisition. All authors met the authorship criteria based on the latest guidelines of the International Committee of Medical Journal Editors. The paper has been seen and approved by all authors.

Acknowledgments

We thank all participants for their participating in this study. In addition, we would also like to express our gratitude to the long-term care facilities that permitted us to conduct this research in their institutions.

References

- Appel, L., Appel, E., Bogler, O., Wiseman, M., Cohen, L., Ein, N., Abrams, H. B., & Campos, J. L. (2020). Older adults with cognitive and/or physical impairments can benefit from immersive virtual reality experiences: A feasibility study. *Frontiers in Medicine*, 6, 329. <https://doi.org/10.3389/fmed.2019.00329>
- Balki, E., Hayes, N., & Holland, C. (2022). Effectiveness of technology interventions in addressing social isolation, connectedness, and loneliness in older adults: Systematic umbrella review. *JMIR Aging*, 5, e40125. <https://doi.org/10.2196/40125>
- Bauer, A. C. M., & Andringa, G. (2020). The potential of immersive virtual reality for cognitive training in elderly. *Gerontology*, 66, 614–623. <https://doi.org/10.1159/000509830>
- Dahlkvist, E., Hartig, T., Nilsson, A., Högberg, H., Skovdahl, K., & Engström, M. (2016). Garden greenery and the health of older people in residential care facilities: A multi-level cross-sectional study. *Journal of Advanced Nursing*, 72, 2065–2076. <https://doi.org/10.1111/jan.12968>
- Drazich, B. F., McPherson, R., Gorman, E. F., Chan, T., Teleb, J., Galik, E., & Resnick, B. (2023). In too deep? A systematic literature review of fully-immersive virtual reality and cybersickness among older adults. *Journal of the American Geriatrics Society*, 71, 3906–3915. <https://doi.org/10.1111/jgs.18553>
- Finnegan, D. J., & Campbell, S. (2023). Tackling loneliness and isolation in older adults with virtual reality: How do we move forward? *Gerontology and Geriatric Medicine*, 9, 23337214231186204. <https://doi.org/10.1177/23337214231186204>
- Gong, K., Wang, C., & Yin, J. (2024). Effects of the natural environment on the subjective and psychological well-being of older people in the community in China. *Buildings*, 14, 2854. <https://doi.org/10.3390/buildings14092854>
- Graneheim, U. H., Lindgren, B. M., & Lundman, B. (2017). Methodological challenges in qualitative content analysis: A discussion paper. *Nurse Education Today*, 56, 29–34. <https://doi.org/10.1016/j.nedt.2017.06.002>
- Healy, D., Flynn, A., Conlan, O., McSharry, J., & Walsh, J. (2022). Older adults' experiences and perceptions of immersive virtual reality: Systematic review and thematic synthesis. *JMIR Serious Games*, 10, e35802. <https://doi.org/10.2196/35802>
- Hennink, M. M., Kaiser, B. N., & Marconi, V. C. (2017). Code saturation versus meaning saturation: How many interviews are enough? *Qualitative Health Research*, 27, 591–608. <https://doi.org/10.1177/1049732316665344>
- Hosseini, M., Thomas, R., Pilutti, L., Fallavollita, P., & Jutai, J. W. (2024). Assessing virtual reality acceptance in long-term care facilities: A quantitative study with older adults. *Disability and Rehabilitation: Assistive Technology*, 19, 2602–2614. <https://doi.org/10.1080/17483107.2023.2295946>
- Jaros, E. (2023). Direct exposure to green and blue spaces is associated with greater mental wellbeing in older adults. *Journal of Aging and Environment*, 37, 460–477. <https://doi.org/10.1080/26892618.2022.2109792>
- Kalantari, S., Xu, T., Mostafavi, A., Kim, B., Dilanchian, A., Lee, A., Boot, W., & Czaja, S. (2023). Using immersive virtual reality to enhance social interaction among older adults: A cross-site investigation. *Innovation in Aging*, 7. <https://doi.org/10.1093/geron/igad031>
- Kaplan, R., & Kaplan, S. (1989). *The experience of nature: A psychological perspective*. Cambridge University Press.
- Kellert, S. R., & Wilson, E. O. (1995). *The biophilia hypothesis* (Vol. 4). Island Press.
- Kim, J. I., & Kim, G. (2025). Exploring subjective mental health change in older adults after experiencing an immersive virtual reality (IVR) -based natural environment. *Journal of Applied Gerontology*, 44, 364–376. <https://doi.org/10.1177/07334648241272042>
- Kolster, A., Rautiainen, L. J., Aalto, U. L., Jansson, A., Partonen, T., Sachs, A. L., Litt, J. S., Masó-Aguado, M., & Pitkälä, K. H. (2025). The importance of nature and wishes for nature-based experiences among older adults in assisted living facilities. *Geriatric Nursing*, 63, 300–306. <https://doi.org/10.1016/j.geri.2025.03.061>
- Lau, K. K.-L., Yung, C. C.-Y., & Tan, Z. (2021). Usage and perception of urban green space of older adults in the high-density city of Hong Kong. *Urban Forestry & Urban Greening*, 64, 127251. <https://doi.org/10.1016/j.ufug.2021.127251>
- Leung, C., Wong, K. C., So, W. W. Y., Tse, Z. C. K., Li, D., Cao, Y., & Shum, D. H. K. (2022). The application of technology to improve cognition in older adults: A review and suggestions for future directions. *Psych Journal*, 11, 583–599. <https://doi.org/10.1002/pchj.565>
- Lincoln, Y. S., & Guba, E. G. (1985). *Naturalistic inquiry*. Sage Publications.
- Mao, Q., Zhao, Z., Yu, L., Zhao, Y., & Wang, H. (2024). The effects of virtual reality-based reminiscence therapies for older adults with cognitive impairment: Systematic review. *Journal of Medical Internet Research*, 26, e53348. <https://doi.org/10.2196/53348>
- McLean, G., Krey, N., & Barhorst, J. B. (2025). Revealing the double-edged sword: Introducing the technology and consumer well-being paradox model. *Psychology & Marketing*, 42, 5–20. <https://doi.org/10.1002/mar.22110>
- Muslu, L., Karakus, Z., Asi, E., Bayindir, R., & Ozer, Z. (2025). Time travel of older people through virtual reality: A qualitative

- study. *BMC Geriatrics*, 25, 42. <https://doi.org/10.1186/s12877-025-05699-x>
- Nikitin, J., Rupprecht, F., Ristl, C., & Korlat, S. (2024). Nature exposure is associated with positive views on aging. *Innovation in Aging*, 8(Supplement_1), 264–265. <https://doi.org/10.1093/geroni/igae098.0857>
- O'Brien, B. C., Harris, I. B., Beckman, T. J., Reed, D. A., & Cook, D. A. (2014). Standards for reporting qualitative research: A synthesis of recommendations. *Academic Medicine*, 89. https://journals.lww.com/academicmedicine/fulltext/2014/09000/standards_for_reporting_qualitative_research__a.21.aspx
- Pfeiffer, E. (1975). A short portable mental status questionnaire for the assessment of organic brain deficit in elderly patients. *Journal of the American Geriatrics Society*, 23, 433–441. <https://doi.org/10.1111/j.1532-5415.1975.tb00927.x>
- Schaumburg, M., Imtiaz, A., Zhou, R., Bernard, M., Wolbers, T., & Segen, V. (2025). Immersive virtual reality for older adults: Challenges and solutions in basic research and clinical applications. *Ageing Research Reviews*, 109, 102771. <https://doi.org/10.1016/j.arr.2025.102771>
- Shah, S. G. S., Nogueras, D., van Woerden, H. C., & Kiparoglou, V. (2021). Evaluation of the effectiveness of digital technology interventions to reduce loneliness in older adults: Systematic review and meta-analysis. *Journal of Medical Internet Research*, 23, e24712. <https://doi.org/10.2196/24712>
- Sia, A., Tam, W. W. S., Fogel, A., Kua, E. H., Khoo, K., & Ho, R. C. M. (2020). Nature-based activities improve the well-being of older adults. *Scientific Reports*, 10, 18178. <https://doi.org/10.1038/s41598-020-74828-w>
- Smalley, A. J., White, M. P., Sandiford, R., Desai, N., Watson, C., Smalley, N., Tuppen, J., Sakka, L., & Fleming, L. E. (2023). Soundscapes, music, and memories: Exploring the factors that influence emotional responses to virtual nature content. *Journal of Environmental Psychology*, 89, 102060. <https://doi.org/10.1016/j.jenvp.2023.102060>
- Steensma, R., van den Bogerd, N., Dijkstra, K., Janssen-Heijnen, M., Krabbendam, L., Vries, R. d., & Maas, J. (2025). How to implement nature-based interventions in hospitals, long-term care facilities for elderly, and rehabilitation centers: A scoping review. *Urban Forestry & Urban Greening*, 103, 128587. <https://doi.org/10.1016/j.ufug.2024.128587>
- Stefan, H., Watkins, V., Hutchinson, A. M., & Horan, B. (2025). Enhancing the daily lives of aged care residents including those with mild cognitive impairment through virtual reality travel experiences. *Virtual reality*, 29, Article 106. <https://doi.org/10.1007/s10055-025-01171-8>
- Tong, K., Thompson, C. W., Carin-Levy, G., Liddle, J., Morton, S., & Mead, G. E. (2025). Nature-based interventions for older adults: A systematic review of intervention types and methods, health effects and pathways. *Age and Ageing*, 54. <https://doi.org/10.1093/ageing/afaf084>
- Ulrich, R. S. (1984). View through a window may influence recovery from surgery. *Science*, 224, 420–421. <https://doi.org/10.1126/science.6143402>
- Ulrich, R. S., Simons, R. F., Losito, B. D., Fiorito, E., Miles, M. A., & Zelson, M. (1991). Stress recovery during exposure to natural and urban environments. *Journal of Environmental Psychology*, 11, 201–230. [https://doi.org/10.1016/S0272-4944\(05\)80184-7](https://doi.org/10.1016/S0272-4944(05)80184-7)
- Whear, R., Coon, J. T., Bethel, A., Abbott, R. A., Stein, K., & Garside, R. (2014). What is the impact of using outdoor spaces such as gardens on the physical and mental well-being of those with dementia? A systematic review of quantitative and qualitative evidence. *Journal of the American Medical Directors Association*, 15, 697–705. <https://doi.org/10.1016/j.jamda.2014.05.013>
- Wilson, E. O. (1984). *Biophilia*. Harvard University Press.
- Wu, Y. T., Prina, A. M., Jones, A., Matthews, F. E., & Brayne, C. (2015). Older people, the natural environment and common mental disorders: Cross-sectional results from the cognitive function and ageing study. *BMJ Open*, 5, e007936. <https://doi.org/10.1136/bmjopen-2015-007936>
- Xiaoxue, S., & Huang, X. (2024). Promoting stress and anxiety recovery in older adults: Assessing the therapeutic influence of biophilic green walls and outdoor view. *Frontiers in Public Health*, 12, 1352611. <https://doi.org/10.3389/fpubh.2024.1352611>
- Yan, T., Leng, H., & Yuan, Q. (2023). The role of “Nostalgia” in environmental restorative effects from the perspective of healthy aging: Taking Changchun parks as an example. *Land*, 12. <https://doi.org/10.3390/land12091817>
- Yao, Y. F., & Chen, K. M. (2017). Effects of horticulture therapy on nursing home older adults in southern Taiwan. *Quality of Life Research*, 26, 1007–1014. <https://doi.org/10.1007/s11136-016-1425-0>
- Yuan, S., Tao, F., & Li, Y. (2022). The restorative effects of virtual reality forests on elderly individuals during the COVID-19 lockdown. *Journal of Organizational and End User Computing*, 34, 22. <https://doi.org/10.4018/JOEUC.297626>

© The Author(s) 2026. Published by Oxford University Press on behalf of the Gerontological Society of America.

This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs licence (<https://creativecommons.org/licenses/by-nc-nd/4.0/>), which permits non-commercial reproduction and distribution of the work, in any medium, provided the original work is not altered or transformed in any way, and that the work is properly cited. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com.

Innovation in Aging, 2026, 10, 1–12

<https://doi.org/10.1093/geroni/igag045>

Original Research Article